

Student Resource Guide

Generative AI for Beginners

Section 6: Training and Fine-Tuning | Lecture 6.1: The Pre-Training Process

Learning Objectives

- Explain what pre-training is and why it is described as 'foundational education' for AI models
- Describe the major data sources used to pre-train large language models and image models
- Explain the self-supervised learning objective used during language model pre-training
- Describe the compute and infrastructure requirements of large-scale pre-training
- Evaluate the practical implications of pre-training data composition for model capability and bias

Introduction

Pre-training is the process that transforms a randomly initialised neural network—essentially a sophisticated random number generator—into a capable AI system with broad knowledge and competence. It is the most computationally intensive, most expensive, and most consequential phase of AI model development. Understanding pre-training allows practitioners to reason about where model capabilities come from, why certain models have specific knowledge strengths and gaps, and what constraints shape the behaviour of commercial AI systems they work with.

The scale of modern pre-training is difficult to comprehend intuitively. Training GPT-3 consumed an estimated 355 GPU-years of compute at a cost of approximately \$4.6 million in cloud compute alone—and that was in 2020. Current frontier models require orders of magnitude more. This scale has profound implications: only a small number of organisations globally can afford to pre-train frontier models, creating a stratified AI ecosystem where most practitioners work with pre-trained models as infrastructure rather than as something they build.

For practitioners who will never train a model from scratch, pre-training knowledge is still directly valuable. It explains why models behave as they do—their knowledge biases, their training cutoffs, their blind spots, and their unexpected capabilities. Understanding the pre-training process is foundational to using pre-trained models effectively, fine-tuning them appropriately, and explaining their outputs accurately to stakeholders.

Key Concepts

Self-Supervised Learning Objective

Definition	The training approach used in language model pre-training where the model generates its own supervision signal from unlabelled data. For autoregressive language models (GPT family), the objective is next-token prediction: given a sequence of tokens, predict the next token. For masked language models (BERT family), some tokens are masked and the model predicts them from context. No human-labelled data is required.
Business Context	Self-supervised learning is what makes pre-training economically feasible at scale—it requires no human annotation. A trillion-token training corpus can be assembled from existing internet text without any labelling cost. This contrasts sharply with supervised learning, where annotation cost grows linearly with dataset size.
AI Practitioner relevance	Practitioners evaluating pre-trained models should understand that the model's capabilities emerge from the statistics of what it learned to predict. A model that frequently saw medical text will understand medical terminology well; one trained predominantly on English internet text will perform poorly on medical texts in Hindi. Capability directly reflects training data distribution.
Real-World Example	GPT-3 was trained on approximately 300 billion tokens using next-token prediction alone. It developed the ability to write code, solve mathematical problems, and answer factual questions as emergent capabilities—none of these were explicitly trained targets. They emerged because predicting the next token in code and factual text requires understanding these domains.

Pre-Training Data Composition

Definition	The specific sources, proportions, and quality filters applied to the training corpus. Typical large language model
-------------------	---

	training data includes web text (Common Crawl), books, Wikipedia, scientific papers, code, and news. Data composition fundamentally determines model capability distribution—what the model knows well and what it knows poorly.
Business Context	Data composition choices create commercial differentiation between foundation models. A model trained with heavy emphasis on code repositories (GitHub) will outperform a general model on coding tasks. A model trained on legal documents will understand legal reasoning better. Organisations choosing between foundation models should consider training data composition, not just model architecture.
AI Practitioner relevance	When a pre-trained model consistently underperforms on domain-specific tasks, data composition is often the root cause. This informs the fine-tuning strategy: domain-specific fine-tuning can correct this gap for task-specific applications; for general domain improvement, a model with better-aligned pre-training data may be preferable.
Real-World Example	Code Llama (Meta) is a Llama 2 base model further pre-trained on a large corpus of code. This additional code-focused pre-training—rather than the model architecture—accounts for its dramatically improved coding performance compared to the base Llama 2 model on the same architecture.
Key Risks	Web-scraped training data contains misinformation, toxic content, personally identifiable information, and copyrighted material. Even after filtering, these contaminants influence model outputs—producing models that may generate harmful content, factual errors, or outputs resembling copyrighted works.

Scaling Laws

Definition	Empirical relationships discovered by Kaplan et al. (2020) and Hoffmann et al. (2022, 'Chinchilla') showing that model performance improves predictably with
-------------------	--

	increases in model parameters, training data volume, and compute budget—following power laws. The Chinchilla scaling laws further established optimal compute allocation between model size and data quantity.
Business Context	Scaling laws enable compute budget planning: given a fixed budget, organisations can predict the expected performance of different model size / data combinations before training. This turns model development from guesswork into engineering. Scaling laws also explain why model quality has improved so reliably as compute costs have fallen.
AI Practitioner relevance	Practitioners evaluating whether to use a larger or smaller model variant should understand that scaling laws predict capability improvements, but not necessarily improvements on every specific task. Smaller models fine-tuned on domain-specific data often outperform larger general models on specific tasks—because task-specific fine-tuning provides stronger signal than general scaling.
Real-World Example	Chinchilla scaling laws revealed that GPT-3 (175B parameters) was undertrained relative to its size—it needed approximately 3.5 trillion tokens for optimal compute efficiency at that parameter count. Meta's Llama models were designed from the start with Chinchilla-optimal data-to-parameter ratios, producing models that outperform larger but undertrained predecessors.

Compute Infrastructure for Pre-Training

Definition	The hardware systems required for large-scale model pre-training: thousands of specialised accelerators (GPUs or TPUs) interconnected with high-bandwidth networking (InfiniBand, NVLink), operating in parallel using distributed training techniques (data parallelism, tensor parallelism, pipeline parallelism) over weeks or months of continuous computation.
Business Context	Pre-training compute costs create

	significant barriers to entry. Training GPT-4-scale models requires clusters of thousands of high-end GPUs running for months—infrastructure accessible only to well-funded organisations. This is why the AI ecosystem has stratified: a small number of organisations train frontier models; the vast majority of practitioners use those models via APIs or open-weight releases.
AI Practitioner relevance	For practitioners considering whether to pre-train a domain-specific model vs. fine-tune an existing one, the compute economics are decisive: fine-tuning a pre-trained model requires 0.01–1% of the compute of pre-training from scratch. Pre-training from scratch is only justified when domain data is so specialised and volume so large that existing pre-trained models cannot be effectively adapted.
Real-World Example	A biomedical AI company evaluated pre-training a domain-specific language model on 500B tokens of biomedical literature vs. fine-tuning Llama 3 on the same data. Fine-tuning required 200 GPU-hours; pre-training would have required 50,000+ GPU-hours. They chose fine-tuning, achieving competitive domain performance at 0.4% of the pre-training compute cost.

Real-World Applications

Pre-training knowledge informs how practitioners select, deploy, and debug foundation models across all AI applications.

Foundation Model Selection for Enterprise Deployment: When selecting a foundation model for enterprise deployment, practitioners should evaluate training data composition alongside benchmarks. A model pre-trained on domain-relevant data will require less fine-tuning effort and produce better baseline performance. Review model cards and technical reports to understand training data sources before committing to a model.

Understanding and Communicating Model Knowledge Cutoffs: Pre-training data has a collection date—information after this date is not in the model's training distribution. Practitioners must communicate knowledge cutoffs to users and implement retrieval-augmented generation (RAG) to supplement with current information where freshness matters. Falsely attributing currency to pre-trained model knowledge is a significant deployment error.

Debugging Model Capability Gaps: When a pre-trained model consistently fails on a category of inputs—specific languages, cultural references, domain terminology, date-sensitive facts—root cause analysis should consider training data composition. Identifying the gap as a pre-training limitation (vs. a prompting or fine-tuning issue) directs the correct remediation strategy.

Responsible AI Assessment: Bias Tracing: Pre-training data composition is the primary source of distributional bias in foundation models. Bias assessment should trace observed model behaviour back to likely training data patterns—which demographic groups are over/under-represented, which cultural contexts are central vs. marginal. This tracing is necessary for responsible AI documentation and bias mitigation planning.

Best Practices

Always read the model card before deploying a pre-trained model. Model cards document training data sources, known biases, intended use cases, and out-of-scope applications. This information is essential for responsible deployment and is often overlooked by practitioners focused only on benchmark performance.

Document and communicate knowledge cutoff dates in all AI-powered products. Users interacting with pre-trained models need to understand that the model's knowledge has a fixed endpoint. Build cutoff disclosure into the user interface and supplement with retrieval systems for time-sensitive queries.

Evaluate domain performance directly, not just benchmark transfer. General benchmarks (MMLU, HumanEval) measure average-case performance. Your specific domain may be over- or under-represented in benchmark test sets. Always evaluate on representative domain examples before deployment.

Do not attempt to pre-train from scratch without justification. The compute and data requirements are prohibitive for most organisations. Fine-tuning and retrieval-augmented generation achieve domain-specific improvements at a fraction of the cost. Reserve pre-training consideration for cases where existing model architecture fundamentally cannot support the required domain.

Common Mistakes to Avoid

Mistake: Assuming a pre-trained model 'knows' everything that existed before its training cutoff.

Correct approach: Pre-training data sampling means the model has differential knowledge density—topics well-represented in its training corpus are known better than topics that appeared rarely. Even facts from well before the cutoff may be unknown if they were absent from training sources.

Mistake: Treating model benchmarks as equivalent to domain performance without validation.

Correct approach: Benchmark scores reflect performance on benchmark-specific test sets, which may not represent your domain. Always validate on representative domain examples before production deployment. A model scoring 90% on MMLU may perform at 60% on domain-specific terminology not well-represented in MMLU.

Mistake: Conflating pre-training with instruction tuning when discussing model capabilities.

Correct approach: Pre-training gives the model knowledge and language understanding. Instruction tuning (RLHF/SFT, a fine-tuning stage) gives the model the ability to follow instructions helpfully and safely. These are distinct phases with distinct purposes. Mixing them up leads to incorrect diagnoses of model failure modes.

Practitioner Tool: Pre-Trained Model Evaluation Scorecard

Type: Assessment Framework

Use this scorecard when evaluating a pre-trained foundation model for a new deployment use case.

Evaluation Criterion	Information Source	Key Questions	Status
Training data sources	Model card / technical report	Which sources? What dates? Any domain data?	
Knowledge cutoff date	Model card	When does knowledge end? Is freshness required?	
Known biases documented?	Model card / bias section	What biases are documented? Relevant to use case?	
Domain benchmark performance	Technical report or self-testing	Performance on domain-relevant tasks?	
Languages supported	Model card	Are all required languages covered at sufficient quality?	
Context window length	Model documentation	Sufficient for longest expected inputs?	
Safety guardrails	Model card / usage policy	What content is filtered? Does this match use case needs?	
Licensing / use rights	License file	Permits intended commercial use?	

Note: Complete this scorecard before procurement or deployment decisions. Re-evaluate when the model vendor releases updated versions.

Knowledge Check

Question 1: A company wants to deploy an AI assistant for customer service that answers questions about events and product launches from the past 6 months. The model they are considering has a knowledge cutoff 8 months ago. What is the correct approach?

- A. Increase the model's inference temperature to allow it to infer recent information
- **B. Implement Retrieval-Augmented Generation (RAG) to supplement the model with current information**
- C. Fine-tune the model on recent news to update its knowledge
- D. Use a larger version of the same model to improve knowledge recency

Correct Answer: B. Implement Retrieval-Augmented Generation (RAG) to supplement the model with current information

Explanation: Pre-trained model knowledge is fixed at training cutoff. RAG retrieves current information from an up-to-date database and provides it as context to the model at inference time—this is the standard approach for knowledge freshness without retraining.

Question 2: A biomedical AI team finds that a general-purpose language model (trained on web text, books, and code) performs poorly on medical terminology extraction. The team is debating between pre-training a new model from scratch and fine-tuning. What should drive this decision?

- A. Pre-training is always better for domain-specific tasks
- **B. Fine-tuning is 0.01–1% of the compute cost—the team should fine-tune unless domain volume and specialisation uniquely justify pre-training**
- C. Only pre-training can improve domain performance; fine-tuning only adjusts style
- D. The decision should be based solely on benchmark performance, not compute cost

Correct Answer: B. Fine-tuning is 0.01–1% of the compute cost—the team should fine-tune unless domain volume and specialisation uniquely justify pre-training

Explanation: Fine-tuning is dramatically more compute-efficient than pre-training and typically achieves competitive domain performance. Pre-training from scratch is only justified when the domain is so specialised that existing pre-trained models cannot be effectively adapted.

Question 3: A model's benchmark score on MMLU (Massive Multitask Language Understanding) is 78%. A practitioner deploys it for legal document summarisation without domain testing. The model performs poorly. What most likely explains this?

- A. The model's context window is too small for legal documents
- **B. MMLU does not specifically test legal reasoning—benchmark scores do not predict domain-specific performance reliably without domain validation**
- C. Legal documents require fine-tuning before any language model can process them
- D. The model's knowledge cutoff predates modern legal terminology

Correct Answer: B. MMLU does not specifically test legal reasoning—benchmark scores do not predict domain-specific performance reliably without domain validation

Explanation: General benchmarks measure average-case performance across diverse topics. Legal reasoning may be underrepresented in MMLU. Domain validation on representative legal examples is required before deployment to verify actual performance.

Reflection Questions

Take time to consider the following questions before the next session:

1. Pre-training at frontier scale is concentrated in a small number of organisations. What are the implications of this concentration for AI access, safety oversight, and competitive dynamics?
2. Self-supervised learning allows models to learn from unlabelled internet text. What quality and bias trade-offs does this create compared to smaller, curated, human-annotated datasets?
3. Scaling laws suggest that more compute reliably improves model quality. At what point might non-compute factors (data quality, architecture innovation, training techniques) become the binding constraint?
4. A model trained on web data inherits the biases, misinformation, and cultural assumptions of internet content. What obligations do deployers of pre-trained models have to assess and mitigate these characteristics?
5. Knowledge cutoffs create a systematic gap between model knowledge and current reality. How should this limitation be communicated to users of AI-powered products?

Key Takeaways

- Pre-training is self-supervised learning on massive datasets—no human labels required. For language models, the objective is next-token prediction.
- Training data composition determines capability distribution: the model knows what is in its training data, in proportion to how much it appeared.
- Scaling laws predict that performance improves predictably with more parameters, data, and compute—enabling principled compute budget planning.

- Pre-training frontier models requires thousands of GPUs running for months at costs of tens to hundreds of millions of dollars, accessible only to a small number of organisations.
- Fine-tuning is 0.01–1% of the compute cost of pre-training—the correct choice for domain adaptation in almost all practical cases.
- Practitioners must understand training data composition, knowledge cutoffs, and documented biases before deploying any pre-trained model.

Next Actions

Complete at least two of the following before the next lecture:

6. Read the model card for one foundation model you currently use (e.g., Claude, GPT-4, Llama 3) and document its training data sources, knowledge cutoff, and listed limitations.
7. Identify one domain-specific task relevant to your work and test the model's performance—compare to its general benchmark score to assess domain vs. general performance alignment.
8. Research the Chinchilla scaling laws paper ('Training Compute-Optimal Large Language Models', Hoffmann et al. 2022) abstract to understand the core insight about data-to-parameter ratios.
9. Evaluate whether a RAG implementation would address knowledge freshness requirements for one AI application in your organisation where the model's training cutoff creates gaps.